

HUMAN ID PREDICTION USING GAIT PATTERN RETRIEVED FROM WI-FI SIGNALS

*Dissertation submitted to
Visvesvaraya National Institute of Technology, Nagpur
in fulfillment of requirement for the award of
degree of*

Master of Technology
in
Applied AI

by
Sanjeev Kumar Pandey (MT23AAI001)

under the guidance of

Dr. Amit Agarwal
Assistant Professor, Department of ECE, VNIT



Department of Electronics and Communication Engineering
Visvesvaraya National Institute of Technology, Nagpur
Nagpur 440010 (India)

December, 2025

HUMAN ID PREDICTION USING GAIT PATTERN RETRIEVED FROM WI-FI SIGNALS

*Dissertation submitted to
Visvesvaraya National Institute of Technology, Nagpur
in fulfillment of requirement for the award of
degree of*

Master of Technology
in
Applied AI

by
Sanjeev Kumar Pandey (MT23AAI001)

under the guidance of

Dr. Amit Agarwal
Assistant Professor, Department of ECE, VNIT



Department of Electronics and Communication Engineering
Visvesvaraya National Institute of Technology, Nagpur
Nagpur 440010 (India)

December, 2025

Copyright © Visvesvaraya National Institute of Technology, Nagpur(VNIT)
December, 2025



DECLARATION

I, "Sanjeev Kumar Pandey" hereby declare that the project work titled "**Human ID Prediction using Gait Pattern retrieved from Wi-Fi Signals**" is carried out by us in the Electronics and Communication Engineering of VNIT, Nagpur, India. The work is original and has not been submitted earlier as a whole or in part for the award of any degree/diploma at this or any other Institution/University.

Sanjeev Kumar Pandey
MT23AAI001

Date: December 10, 2025

CERTIFICATE

This is to certify that the dissertation titled "**Human ID Prediction using Gait Pattern retrieved from Wi-Fi Signals**" submitted by **Sanjeev Kumar Pandey** in partial fulfilment of requirements for the award of Master of Technology in the Applied AI of VNIT, Nagpur. The work is comprehensive, complete and fit for evaluation.

Dr. Amit Agarwal
Assistant Professor,
Electronics and Communication Engineering,
VNIT, Nagpur

Dr. Ashwin Kothari
Professor and Head,
Electronics and Communication Engineering, VNIT, Nagpur
Date: December 10, 2025

ACKNOWLEDGEMENTS

Our sincere gratitude to our guide, for his constant support and encouragement, without which the project would not have been possible. Our heartfelt gratitude to him for offering all of the facilities and assisting us in overcoming obstacles.

We would like to thank all the professors of ECE department for helping us by some way or the other. We sincerely thank our project evaluators for their valuable comments and suggestions . We thank Head of ECE department for giving an opportunity for the project.

Date: December 10, 2025

Sanjeev Kumar Pandey (MT23AAI001)

Abstract

The rapid spread of indoor Wi-Fi has turned everyday spaces into powerful, passive sensing systems—no wearables, no cameras, just the invisible radio waves already around us. From these signals, Channel State Information (CSI) reveals fine-grained distortions caused by human movement, exposing distinctive gait patterns that are extremely hard to fake or imitate. This dissertation harnesses Wi-Fi CSI for Human Identification (Human Id), rigorously exploring deep learning architectures designed to handle rich, multivariate time-series CSI data along with auxiliary metadata to push the limits of reliable, device-free biometric recognition.

We built and tested our models on a rich, publicly available benchmark of CSI traces, captured from 30 subjects performing 5 distinct experiments in both line-of-sight and non-line-of-sight indoor settings. For Gait Pattern, Experiment#3 is selected having 4 different walking activities. Four cutting-edge deep learning architectures—DenseNet1D, EfficientNet1D-LSTM, MobileNetV3-1D-LSTM, and a custom CSILSTMNet—were all trained within a unified, production-grade MLOps pipeline, leveraging MLflow for end-to-end experiment tracking, systematic hyperparameter optimization, and rock-solid reproducibility.

We systematically pushed the limits of generalization by sweeping learning rates, batch sizes, optimizers, and weight-decay settings. DenseNet1D emerged as the clear winner: with a learning rate of 0.0001, batch size 64, and Adam with 1e-05 weight decay, it delivered the highest validation accuracy (up to 98.33%) and F1-score (Validation 98.32%), decisively outperforming all other architectures. MobileNetV3-1D-LSTM and CSILSTMNet also posted strong, competitive results when optimally tuned.

This study exposes how sharply CSI-based identification performance hinges on architectural design and training choices—small tweaks can have big consequences. It further proves that CSI is a powerful, truly passive, and non-intrusive biometric signal, ideal for secure indoor monitoring, with clear potential in high-security facilities, smart buildings, and healthcare settings. Overall, this dissertation delivers a rigorous, end-to-end assessment of deep learning methods for Wi-Fi-based HAR and Human-ID, pinpoints the best-performing architecture, and cements an MLOps-driven framework for fast, scalable, and reproducible experimentation.

List of Figures

3.1	System architecture for CSI-based gait identification.	13
3.2	Overall WiFi CSI-based Human Identification Pipeline. This diagram illustrates the high-level flow of configuration loading, dataset handling, preprocessing, model selection, and the training-validation loop used in this research.	14
3.3	Detailed Training and Model Selection Workflow. This slide illustrates the dataset preprocessing, model selection across four architectures, and the training-validation loop executed for every experiment using the MLOps-enabled pipeline.	15
3.4	Compact overview of the CSI preprocessing workflow. Raw complex CSI is decomposed into magnitude and phase, the phase channel is sanitised, and both components are concatenated to form the final CSI tensor used for model training.	18
3.5	CSILSTMNet architecture.	19
3.6	DenseNet1D architecture.	20
3.7	EfficientNet1DLSTM architecture.	20
3.8	MobileNetV3-1D-LSTM architecture.	20
3.9	MLflow-based comparison of the four CSI models across training and validation accuracy, precision, F1-score, and loss curves over 150 epochs. These metrics were automatically logged by MLflow, enabling seamless model comparison and reproducible performance tracking.	22
a	Training accuracy	22
b	Training precision	22
c	Training F1-score	22
d	Validation accuracy	22
e	Validation precision	22
f	Validation F1-score	22
g	Training loss	22
h	Validation loss	22

4.1	Validation accuracy comparison of 4 models over epochs using MLFlow	26
4.2	Validation precision comparison of 4 models over epochs using MLFlow	26
4.3	Validation F1-score comparison of 4 models over epochs using MLFlow	26
4.4	CSILSTMNet (14 Nov): training and validation accuracy/loss over epochs.	27
4.5	MobileNetV3-1D-LSTM (14 Nov): training and validation accuracy/loss over epochs.	27
4.6	EfficientNet1D-LSTM (14 Nov): training and validation accuracy/loss over epochs.	28
4.7	DenseNet1D (14 Nov): training and validation accuracy/loss over epochs.	28
4.8	CSILSTMNet (14 Nov): confusion matrix on the validation split. . .	29
4.9	MobileNetV3-1D-LSTM (14 Nov): confusion matrix on the validation split.	29
4.10	EfficientNet1D-LSTM (14 Nov): confusion matrix on the validation split.	30
4.11	DenseNet1D (14 Nov): confusion matrix on the validation split. . . .	30
4.12	Best DenseNet1D configuration (27 Nov): training and validation accuracy/loss.	31
4.13	Confusion matrix for the best DenseNet1D model (27 Nov).	32
6.1	Graphical abstract of the CSI-based gait identification pipeline, illustrating sensing, data processing, model comparison, MLOps integration, and downstream applications.	41

List of Tables

1.1	Experiments and activities indicators in captured Wi-Fi packets for the dataset	6
1.2	Data files naming convention	6
1.3	Details of the utilized environments along with the subject's gender, age, weight, and height.	7
3.1	Models Used for Training: Overview	19
4.1	14 November — Performance comparison of four deep learning architectures.	25
4.2	27 November — DenseNet1D hyperparameter sweep results.	31
5.1	Hardware Configuration Used for Model Training and Validation . . .	36
5.2	Key Software and Library Versions	37
5.3	Summary of Important Source Files in the phase2 Directory	38

Contents

Abstract	i
List of Figures	i
List of Tables	iii
1 Introduction	1
1.1 Introduction	1
1.1.1 Background and Motivation	1
1.1.2 Research Context and Dataset Overview	2
1.1.3 Role of MLOps and Experimentation Strategy	2
1.1.4 Technical Rationale and Deep Learning Perspective	2
1.1.5 Critical Analysis of Existing Limitations	3
1.1.6 Contributions of This Thesis	3
1.1.7 Chapter Summary	4
1.2 Dataset Description	4
1.2.1 Overview	4
1.2.2 Data Collection Setup	4
1.2.3 Dataset Structure	5
1.2.4 Activities Covered	5
1.2.5 Participant Details	6
1.2.6 Data Volume Summary	7
1.2.7 Suitability for This Research	7
2 Literature Review - Classification	9
2.1 Literature Review	9
2.1.1 Foundations of Wi-Fi CSI-Based Human Sensing	9
2.1.2 Deep Learning Architectures for CSI Interpretation	9
2.1.3 Privacy-Preserving Sensing and Biometric Identification	10

2.1.4	Challenges in CSI Data: Noise, Environment, and Variability .	10
2.1.5	Comparative Model Evaluations and Benchmarking Efforts . .	10
2.1.6	Critical Analysis of Existing Work	11
2.1.7	Motivation for the Present Study	12
3	Methodology	13
3.1	Overall System Architecture	13
3.2	CSI Data and Preprocessing Pipeline	15
3.2.1	Complex CSI Decomposition: Magnitude and Phase	16
3.2.2	Phase Sanitization	16
3.2.3	Final CSI Feature Tensor	17
3.2.4	CSI Preprocessing Diagram	18
3.2.5	Sliding-Window Segmentation	18
3.2.6	Dual-Branch Input Construction	18
3.3	Model Architectures	19
3.3.1	CSILSTMNet	19
3.3.2	DenseNet1D Architecture Diagram	20
3.3.3	EfficientNet1D-LSTM Diagram	20
3.3.4	MobileNetV3-1D-LSTM Diagram	20
3.4	Training Process	20
3.5	MLflow-Based MLOps Integration	21
3.5.1	Experiment Tracking and Automated Logging	21
3.5.2	Why MLflow Integration Matters	21
3.5.3	Outcome	22
3.6	Evaluation Metrics	22
4	Results and Discussions	24
4.1	Results	24
4.1.1	Overall Performance Overview	24
4.1.2	14 November: Four-Model Comparative Analysis	25
4.1.3	27 November: DenseNet Hyperparameter Optimisation	31
4.1.4	Combined Interpretation and Statistical Insights	32
4.2	Discussion	33
4.2.1	Why DenseNet1D Outperforms Competing Architectures . . .	33
4.2.2	Role of Hybrid CNN–LSTM Models	34
4.2.3	Limitations of LSTM-Only Approaches	34
4.2.4	Impact of CSI Variability and Environmental Factors	34

4.2.5	Implications for Real-World Deployment	35
4.2.6	Summary	35
5	Hardware and Software Environment and Source Code	36
5.1	Hardware Specifications	36
5.2	Software Environment	36
5.3	Environment Summary	37
5.4	Project Source Code and Repository Structure	37
6	Summary and Conclusions	39
6.0.1	Summary	39

Chapter 1

Introduction

1.1 Introduction

1.1.1 Background and Motivation

As Wi-Fi infrastructure has become nearly ubiquitous in modern indoor environments, it has evolved from a communication utility into a versatile medium for passive sensing. This capability enables advanced applications such as human activity recognition (HAR) and gait-based individual identification, offering a more secure and privacy-preserving alternative to conventional biometric systems that can be easily spoofed or visually manipulated [17].

The foundation of these capabilities lies in the unique properties of Wi-Fi Channel State Information (CSI), which encodes minute perturbations in the wireless channel caused by human motion. These subtle variations are extremely difficult to replicate or forge, rendering CSI-based biometric identification inherently robust against impersonation attacks [2]. Moreover, because CSI captures fine-grained dynamics related to body posture, stride, and micro-movements, it exhibits strong discriminability across individuals, enabling precise gait modelling [31].

Importantly, Wi-Fi-based sensing is passive, device-free, and non-intrusive, making it suitable for environments where privacy, stealth, or minimal hardware deployment are critical. Applications extend across security monitoring, smart ambient environments, health-care analytics, and continuous authentication [13]. The fact that these systems operate using existing Wi-Fi routers further enhances scalability and cost-efficiency.

1.1.2 Research Context and Dataset Overview

This thesis investigates a deep learning-based framework for individual identification through gait analysis using CSI data. The evaluation focuses on the comparative performance of four state-of-the-art architectures: CSILSTMNet, DenseNet1D, EfficientNet1D-LSTM, and MobileNetV3_1D_LSTM [9].

The research leverages a dataset containing recordings from 30 participants performing 12 distinct activities. For the experiments conducted in this work, Experiment #3—comprising exclusively gait-related CSI sequences totalling 1.94 GB—was selected due to its focus on walking patterns essential for biometric identification [1].

1.1.3 Role of MLOps and Experimentation Strategy

A central emphasis of this study is the integration of MLOps practices to ensure reproducibility, standardized experimentation, and efficient comparative evaluation. Using MLflow, all model training runs, hyperparameter combinations, performance metrics, and learned artifacts are tracked systematically, enabling robust cross-architecture comparisons and facilitating transparent model selection [7]. This pipeline allows the identification of the most effective framework based not only on accuracy but also on training stability, generalization capability, and computational efficiency.

Prior work on CSI-based HAR typically reports accuracies between 80–95% when evaluated under controlled lab settings [32]. This thesis extends these investigations by evaluating performance across a broader and more diverse activity set and participant group, thereby providing insights more reflective of real-world deployment scenarios [18].

1.1.4 Technical Rationale and Deep Learning Perspective

The proposed methodology capitalizes on the rich amplitude and phase representations embedded within CSI streams. These features encode gait-specific biometric signatures, representing a substantial advancement beyond binary presence detection or simple classification frameworks [4]. Deep learning architectures are particularly well-suited for this task because they automatically learn hierarchical representations of CSI signals, capturing nonlinear relationships that correlate strongly with human-specific motion traits [20].

By systematically comparing architectures of varying depth, width, and hybrid composition, this research addresses a critical gap in existing literature: the lack of rigorous architectural analysis in CSI-based biometric identification. Earlier studies

demonstrated the feasibility of applying deep learning to CSI signals but seldom evaluated how architectural variations influence the extraction of gait-specific features or affect real-time deployability [20]. This thesis directly targets this research void.

1.1.5 Critical Analysis of Existing Limitations

While CSI-based sensing has demonstrated tremendous potential, the field still faces several constraints that motivate deeper investigation:

- **Limited Architectural Comparisons:** Studies often focus on a single architecture without evaluating how alternative designs may better capture CSI dynamics.
- **Lack of MLOps Integration:** Many prior works lack systematic experimentation pipelines, making reproducibility and hyperparameter analysis difficult.
- **Environmental Constraints:** CSI signals are highly sensitive to router placement, wall materials, multipath effects, and human proximity, yet many works overlook these deployment-specific variations.
- **Dataset Diversity:** Several datasets are collected in narrow or controlled spaces, limiting generalizability across different building layouts.
- **Computational Inefficiency:** Complex deep learning models may achieve high accuracy but cannot be deployed on edge or real-time platforms.

These gaps underscore the need for a comparative, MLOps-driven exploration of multiple architectures under consistent settings—precisely the approach adopted in this thesis.

1.1.6 Contributions of This Thesis

The key contributions of this work can be summarized as follows:

1. A systematic comparative evaluation of four deep learning models for CSI-based gait identification using a multi-participant dataset.
2. An MLOps-driven experimental workflow leveraging MLflow for reproducibility, versioning, and transparent model comparison.
3. Detailed analysis of model performance with respect to architectural depth, hybrid design elements, and CSI signal characteristics.

4. Insights into the suitability of different architectures for real-time and resource-constrained deployments.
5. A refined, deployable approach for privacy-preserving biometric authentication using everyday Wi-Fi infrastructure.

1.1.7 Chapter Summary

This chapter introduced the core motivations behind leveraging Wi-Fi CSI for gait-based biometric identification, highlighting the limitations of traditional biometrics and the advantages of CSI as a passive and privacy-aware modality. It outlined the dataset, architectural focus, and importance of integrating MLOps for systematic experimentation. Additionally, it provided a critical assessment of existing research gaps—particularly the lack of architectural comparisons, limited dataset diversity, and insufficient reproducibility—which directly motivate the objectives of this thesis. The next chapter expands on these foundations by presenting a comprehensive review of related literature across CSI sensing, deep learning architectures, signal processing techniques, and biometric identification frameworks.

1.2 Dataset Description

1.2.1 Overview

This study utilizes a publicly available Wi-Fi Channel State Information (CSI) dataset designed for both Human Activity Recognition (HAR) and Human Identification (Human-ID). The dataset comprises CSI and Received Signal Strength Indicator (RSSI) measurements collected from **30 subjects** performing **12 distinct daily activities** in controlled indoor environments. Each CSI capture provides fine-grained amplitude and phase information across multiple subcarriers, enabling the extraction of biometric gait signatures and motion-related patterns essential for classification tasks[1].

1.2.2 Data Collection Setup

The data was recorded using a standard Wi-Fi transmitter–receiver pair configured to extract CSI at the subcarrier level. The experimental setup includes both:

- **Line-of-Sight (LOS)**

- **Non-Line-of-Sight (NLOS)**

conditions to simulate realistic indoor environments. CSI measurements were captured using a Wi-Fi Network Interface Card (NIC), such as the Intel 5300, capable of logging per-packet CSI information. Each CSI frame contains amplitude and phase values across multiple antenna pairs and subcarriers, along with RSSI and timestamp metadata[1].

1.2.3 Dataset Structure

The dataset is organized by subject and activity, where each recording session is stored as a CSV file containing:

- CSI amplitude values
- CSI phase values
- RSSI measurements
- Timestamps
- Metadata fields (Activity ID, Subject ID, LOS/NLOS)

Each file consists of thousands of CSI packets, typically with:

- 30–60 subcarriers per packet
- Multiple transmit–receive antenna combinations
- High-frequency temporal sampling suitable for deep learning-based time-series modeling

This rich structure enables the use of window-based segmentation (e.g., 64 or 128 samples per window with overlap), which is essential for training deep learning models.

1.2.4 Activities Covered

The dataset includes recordings of **12 daily activities**, encompassing a mixture of static, dynamic, and transitional movements. This diversity supports a comprehensive evaluation of model robustness across multiple behavioral patterns, as detailed in Table 1.1.

This thesis specifically focuses on Experiment 3 (highlighted with a yellow background in the table) for capturing the gait patterns of participants.

Table 1.1: Experiments and activities indicators in captured Wi-Fi packets for the dataset

Experiment indicator	Experiment	Activity indicator	Description
C1	Falling from sitting position	A1	Sit still on a chair
		A2	Falling down
		A3	Lie down
C2	Falling from standing position	A4	Stand still
		A5	Falling down
		A3	Lie down
C3	Walking	A6	Walking from transmitter to receiver
		A7	Turning
		A8	Walking from receiver to transmitter
		A9	Turning
C4	Sit down and stand up	A1	Sit still on a chair
		A10	Standing up
		A4	Stand still
		A11	Sitting down
C5	Pick a pen from the ground	A12	Pick a pen from the ground

Table 1.2: Data files naming convention

Symbol	Abbreviation for	Range
E	Environment	{1, 2, 3}
S	Subject	{1, 2, 3 ..., 30}
C	Experiment Class	{1, 2, 3, 4, 5}
A	Activity	{1, 2, 3, ..., 12}
T	Trial	{1, 2, 3, ..., 20}

1.2.5 Participant Details

A total of 30 participants contributed to the dataset, offering diversity in:

- Gait styles
- Body structure (height, weight)
- Motion characteristics

Each subject performed all 12 activities multiple times, providing sufficient training and testing samples for deep learning-based biometric identification.

Table 1.3: Details of the utilized environments along with the subject’s gender, age, weight, and height.

Environment ID	Environment Configuration	Subject ID	Gender	Age (years)	Weight (kg)	Height (cm)
1	LOS	S1	M	35	75	176
		S2	M	21	63	188
		S3	M	25	75	184
		S4	M	23	70	178
		S5	M	21	77	176
		S6	M	20	85	180
		S7	M	21	90	185
		S8	M	21	115	187
		S9	M	21	90	174
2	LOS	S11	M	23	93	180
		S12	M	26	60	178
		S13	M	20	98	188
		S14	M	21	104	172
		S15	M	25	79	175
		S16	M	21	70	182
		S17	M	24	130	185
		S18	F	24	60	160
		S19	M	23	85	175
		S20	F	24	62	153
1	NLOS	S21	M	21	70	180
		S22	M	22	94	186
		S23	M	21	91	192
		S24	M	21	57	173
		S25	M	21	118	167
		S26	M	21	73	178
		S27	M	21	56	173
		S28	M	23	72	171
		S29	M	23	92	188
		S30	M	21	72	187

1.2.6 Data Volume Summary

Due to repeated activity sessions across all subjects, the dataset includes:

- Hundreds of CSI recordings
- Millions of CSI packets
- Several gigabytes of raw data

This volume supports the training of deep learning architectures that require large-scale time-series input.

1.2.7 Suitability for This Research

The dataset is highly suitable for CSI-based HAR and Human-ID tasks because:

- CSI captures micro-movement signatures unique to each subject.
- LOS and NLOS conditions enable realistic performance evaluation.
- The dataset structure supports windowing and multivariate time-series modeling.
- The diversity of activities promotes generalization across motion patterns.

Thus, this dataset facilitates a rigorous comparison of multiple deep learning architectures for biometric identification and activity classification using Wi-Fi CSI.

Chapter 2

Literature Review - Classification

2.1 Literature Review

Wi-Fi Channel State Information (CSI) has emerged as a powerful medium for passive human sensing, enabling applications in human activity recognition (HAR) and individual identification. This section reviews foundational and recent contributions to CSI-based sensing, focusing on deep learning methodologies, dataset advancements, signal processing innovations, environmental considerations, and comparative model evaluations that motivate the present study.

2.1.1 Foundations of Wi-Fi CSI-Based Human Sensing

Early research demonstrated that the rich spatio-temporal patterns in CSI signals can be effectively leveraged for human activity recognition and user identification [26, 15]. These studies utilized deep learning architectures such as BiLSTMs, CNN-GRU hybrids, InceptionTime, and classical LSTMs to extract discriminative motion signatures embedded within CSI amplitude and phase dynamics [19, 32]. The availability of multi-subject, multi-environment datasets further strengthened the capacity of these models to generalize across subjects and activities [1, 24].

2.1.2 Deep Learning Architectures for CSI Interpretation

Modern CSI-based HAR and identification systems increasingly rely on hybrid and hierarchical architectures. Models combining 1D CNN blocks with global pooling, temporal encoders such as BiLSTM layers, and lightweight yet expressive blocks such as MBConv1D or MobileNetV3 modules have significantly improved feature extraction performance.

These architectures are designed to capture both local spectral variations and long-range temporal dependencies inherent in CSI data. Their success stems from their ability to detect subtle patterns in wireless signal perturbations caused by human gait, gestures, or routine activities.

2.1.3 Privacy-Preserving Sensing and Biometric Identification

CSI-based sensing offers an inherently privacy-preserving alternative to camera-based biometrics. Unlike visual systems, Wi-Fi signals do not capture identifiable visual features; instead, they infer motion patterns from ambient signal distortions [14]. Studies reveal that gait signatures extracted from CSI are extremely difficult to reproduce or spoof, making them trustworthy for biometric authentication [28, 30, 8].

The sensitivity of CSI to fine-grained body movements makes it a strong candidate for robust, unobtrusive human identification, particularly in secure or privacy-sensitive environments.

2.1.4 Challenges in CSI Data: Noise, Environment, and Variability

CSI measurements are inherently influenced by multipath fading, environmental dynamics, network configurations, and hardware variability. This sensitivity necessitates advanced architectures capable of filtering noise and extracting resilient features [10]. Prior research highlights the role of robust signal processing methods—such as amplitude denoising, CSI ratio computation, and phase sanitization—in improving feature stability [25, 27].

Additionally, the reliability of CSI signals is heavily dependent on router placement, environmental layout, and channel settings [32, 10]. Understanding these dependencies is essential for developing generalizable systems suitable for deployment across diverse indoor environments.

2.1.5 Comparative Model Evaluations and Benchmarking Efforts

Several studies emphasize the importance of comparative evaluations across different architectures to identify optimal model choices under varying constraints [24]. Deep

learning models such as DenseNet1D, EfficientNet1D-LSTM, and CSI-LSTMNet have demonstrated promising results in capturing distinctive gait signatures from CSI streams [30].

However, model performance can depend significantly on network characteristics, training strategies, and hyperparameter settings. For this reason, recent benchmarking initiatives advocate open-access datasets and shared codebases to promote transparency, reproducibility, and fair comparison across research efforts [3, 24].

2.1.6 Critical Analysis of Existing Work

Despite significant advancements, several challenges remain in the deployment of CSI-based biometric systems:

Lack of Standardization in Data Collection. Most prior studies employ unique data collection setups, varying in router models, antenna configurations, spacing, and channel properties. This inconsistency complicates cross-paper comparisons and limits generalization across environments.

Overfitting to Controlled Environments. A considerable portion of existing work is evaluated in highly controlled laboratory settings. While impressive accuracies are often achieved, real-world environments involve greater noise, dynamic obstacles, and unpredictable movement, which many models fail to fully address.

Limited Exploration of Multi-Task and Multi-Modal Learning. Although hybrid CNN-LSTM and attention-enhanced networks are widely explored, multi-modal fusion (e.g., CSI + inertial sensors) or multi-task frameworks (e.g., HAR + identification jointly) are rarely studied despite their potential to improve robustness.

Insufficient Focus on Computational Efficiency. Only a few studies evaluate the feasibility of real-time deployment. Lightweight architectures such as MobileNetV3_1D_LSTM remain underexplored despite their suitability for embedded or IoT platforms.

Need for Larger and More Diverse Datasets. Current datasets are smaller and less diverse than those used in other biometric fields (e.g., face recognition). Larger, regionally diverse, multi-floor, multi-AP datasets are essential to reflect realistic deployment scenarios.

These limitations motivate the need for comprehensive comparative studies—such as the one undertaken in this thesis—that evaluate multiple architectures under consistent settings, assess robustness to environmental changes, and consider computational efficiency alongside accuracy.

2.1.7 Motivation for the Present Study

The gaps identified in prior research underscore the necessity of a systematic comparison of lightweight and heavyweight deep learning architectures for CSI-based gait identification. This thesis addresses these challenges by evaluating four architectures—CSILSTMNet, DenseNet1D, EfficientNet1D-LSTM, and MobileNetV3_1D_LSTM on a custom multi-activity, multi-subject dataset. The goal is to determine which model most effectively captures individual-specific movement signatures while maintaining computational efficiency for real-time applications [24, 11].

Furthermore, by integrating standardized preprocessing, controlled training strategies, and environment-specific analyses, this study contributes to the broader pursuit of reliable, privacy-preserving, and scalable CSI-based biometric systems.

Chapter 3

Methodology

This chapter details the complete methodological framework used for developing, training, and evaluating deep learning models for gait-based individual identification using Wi-Fi Channel State Information (CSI). The methodology encompasses data preprocessing, sliding-window sequence preparation, dual-stream feature extraction, model architecture design, MLOps-driven experimentation, and evaluation protocols. To enhance clarity, this chapter integrates system architecture diagrams and model-specific computational flows.

3.1 Overall System Architecture

Figure 3.1 shows the high-level architecture of the proposed CSI-based identification pipeline.

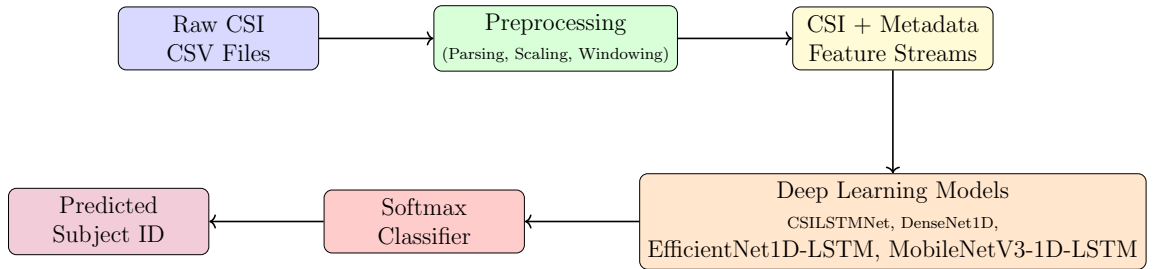


Figure 3.1: System architecture for CSI-based gait identification.

As shown in Figure 3.2, the pipeline begins by loading configurations and preparing the dataset...

WiFi CSI-based Human ID Pipeline

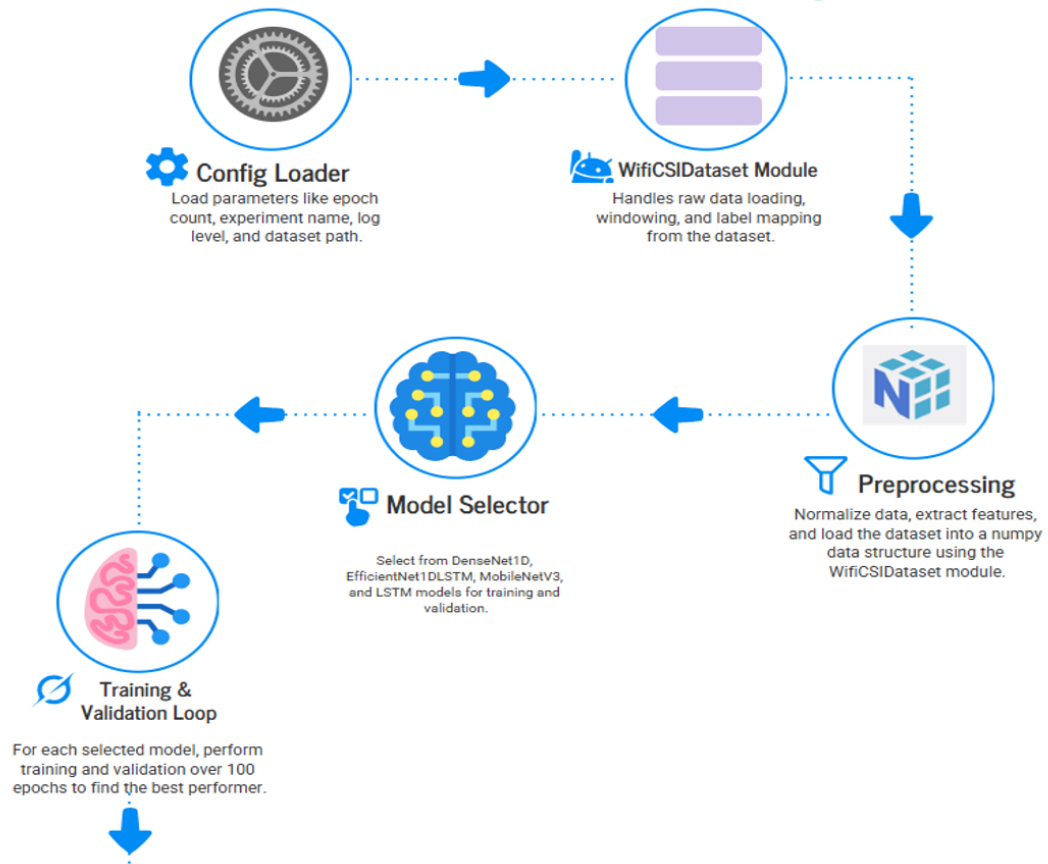


Figure 3.2: Overall WiFi CSI-based Human Identification Pipeline. This diagram illustrates the high-level flow of configuration loading, dataset handling, preprocessing, model selection, and the training-validation loop used in this research.

Figure 3.3 further illustrates the internal training and validation loop...



Figure 3.3: Detailed Training and Model Selection Workflow. This slide illustrates the dataset preprocessing, model selection across four architectures, and the training-validation loop executed for every experiment using the MLOps-enabled pipeline.

3.2 CSI Data and Preprocessing Pipeline

Wi-Fi Channel State Information (CSI) encapsulates the fine-grained propagation characteristics of a wireless link. Each CSI entry is a complex-valued quantity captured across multiple OFDM subcarriers, and therefore contains rich amplitude and phase signatures that respond sensitively to human motion. Before these measurements can be used for model training, they must be decoded, sanitised, and transformed into a structured feature representation. The overall preprocessing workflow is illustrated in Fig. 3.4 and comprises two essential stages: (i) extracting physically meaningful features from raw complex CSI, and (ii) constructing fixed-length temporal windows suitable for deep learning.

3.2.1 Complex CSI Decomposition: Magnitude and Phase

Each CSI value recorded by the Intel 5300 NIC is a complex number $h_i(t) = a_i(t) + j b_i(t)$ for subcarrier i at time t . Rather than discarding the phase—as is commonly done when only `np.abs()` is used—this work preserves both the *magnitude* and the *raw phase* components to retain the complete channel information.

Magnitude. The magnitude reflects the attenuation experienced by the signal and is computed for every subcarrier as:

$$|h_i(t)| = \sqrt{a_i(t)^2 + b_i(t)^2}.$$

This corresponds directly to the `np.abs(z)` operation in NumPy.

Phase. The phase of the CSI conveys path-length variations, micro-Doppler shifts, and fine-grained perturbations caused by human gait. It is extracted as:

$$\theta_i(t) = \text{atan2}(b_i(t), a_i(t)),$$

which matches `np.angle(z)` in the preprocessing script.

Both magnitude and phase are extracted for every CSI packet, producing:

$$X_{\text{mag}} \in \mathbb{R}^{T \times S}, \quad X_{\text{raw_phase}} \in \mathbb{R}^{T \times S},$$

where T is the number of packets and $S = 99$ is the number of subcarriers.

Metadata such as RSSI, AGC, antenna permutation, and link parameters are stored independently in:

$$X_{\text{meta}} \in \mathbb{R}^{T \times M}.$$

3.2.2 Phase Sanitization

Raw CSI phase values are noisy due to:

- phase wrapping at $\pm\pi$,
- carrier-frequency offset (CFO),
- sampling-frequency offset (SFO),
- slowly varying oscillator drift.

To stabilise the phase channel and highlight motion-induced variations, two operations are applied independently to each subcarrier:

- **Unwrapping:** removes 2π discontinuities to create a smooth temporal evolution;
- **Detrending:** subtracts low-frequency drift components, isolating the high-frequency variations attributable to human movement.

The resulting sanitised phase matrix is:

$$X_{\text{phase}} = \text{sanitize_phase}(X_{\text{raw_phase}}).$$

3.2.3 Final CSI Feature Tensor

The model requires a stable, real-valued representation. Therefore, magnitude and sanitised phase are concatenated channel-wise to form the final CSI feature tensor:

$$X_{\text{csi}}(t) = \left[|h_1(t)|, \dots, |h_S(t)|, \tilde{\theta}_1(t), \dots, \tilde{\theta}_S(t) \right] \in \mathbb{R}^{2S}.$$

For $S = 99$ subcarriers, the final dimensionality becomes:

$$X_{\text{csi}} \in \mathbb{R}^{T \times 198}.$$

This dual-channel encoding retains the full physical richness of the wireless channel while suppressing hardware artefacts, thereby improving the deep learning model's ability to identify subtle gait signatures.

Minimal Pseudocode for Feature Extraction

```
for each row in CSI CSV:
    z = parse_complex(row[csi_k])
    mag    = np.abs(z)          # sqrt(a^2 + b^2)
    phase  = np.angle(z)        # atan2(b, a)
    append to X_mag, X_raw_phase

X_phase = sanitize_phase(X_raw_phase)
X_csi   = concatenate(X_mag, X_phase)
```

3.2.4 CSI Preprocessing Diagram

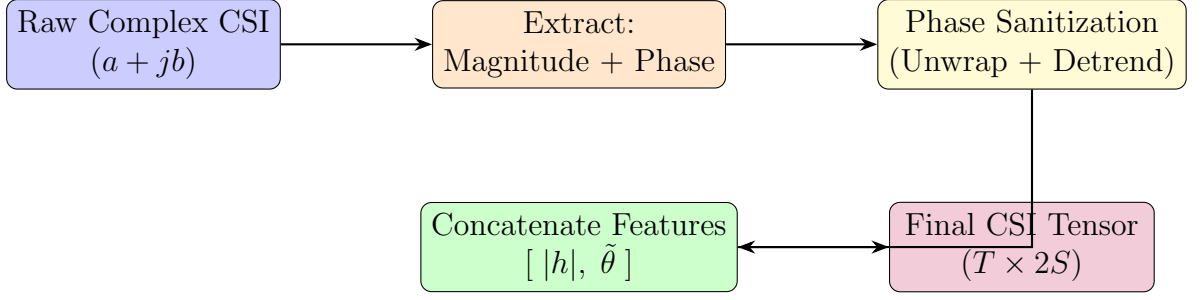


Figure 3.4: Compact overview of the CSI preprocessing workflow. Raw complex CSI is decomposed into magnitude and phase, the phase channel is sanitised, and both components are concatenated to form the final CSI tensor used for model training.

3.2.5 Sliding-Window Segmentation

Since CSI streams are long, noisy, and strongly influenced by human motion dynamics, they are segmented into fixed-size windows. A sliding-window strategy with window length 128 and stride 64 is employed:

- increases sample density,
- preserves temporal continuity,
- ensures coverage over complete gait cycles.

Each CSI window is therefore represented as:

$$X_{\text{CSI}} \in \mathbb{R}^{128 \times 180}, \quad X_{\text{META}} \in \mathbb{R}^{128 \times 12},$$

where the CSI tensor captures subcarrier-wise temporal evolution, and the metadata tensor encodes link-level temporal variation.

3.2.6 Dual-Branch Input Construction

The final preprocessed dataset consists of synchronized CSI and metadata sequences. These are consumed by dual-branch deep learning architectures, where:

- the CSI branch (DenseNet1D, EfficientNet1D-LSTM, MobileNetV3-1D-LSTM) learns fine-grained spatial-temporal patterns from amplitude sequences,
- the metadata branch (LSTM-based) models link-level dynamics that complement CSI features.

This structured transformation—from complex CSI samples to windowed amplitude-metadata sequences—establishes a clean, robust input pipeline supporting accurate gait-based human identification models.

3.3 Model Architectures

Table 3.1: Models Used for Training: Overview

Model Name	CSI Processing Core	Metadata Processing Core	Key Features
CSILSTMNet	Standard LSTM	Standard LSTM	Uses two parallel LSTMs (one for CSI and one for metadata). The final hidden states are concatenated to form the joint representation used for classification.
DenseNet1D	Dense Block 1D CNN	Bidirectional LSTM	Uses 1D CNN blocks (DenseBlock1D) followed by global pooling for CSI, combined with a Bi-LSTM for metadata.
EfficientNet1DLSTM	MBConv1D Blocks	Bidirectional LSTM	Leverages Mobile Inverted Bottleneck (MBConv) blocks (stem, mbconv1-3) designed for efficiency, combined with a Bi-LSTM.
MobileNetV3_1D_LSTM	MobileNetV3 Block 1D	Bidirectional LSTM	Incorporates specialized blocks (including depthwise separable convolutions and Squeeze-and-Excitation (SE) layers) optimized for mobile/efficient computing, combined with a Bi-LSTM.

3.3.1 CSILSTMNet

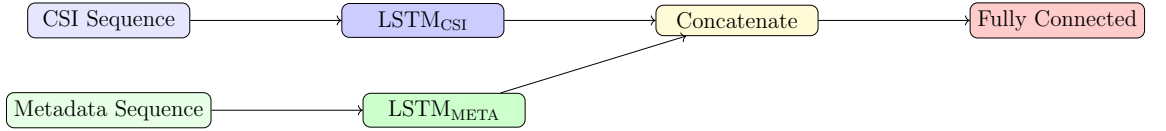


Figure 3.5: CSILSTMNet architecture.

A dual-LSTM structure extracts temporal features independently from CSI and metadata, which are later fused.

3.3.2 DenseNet1D Architecture Diagram

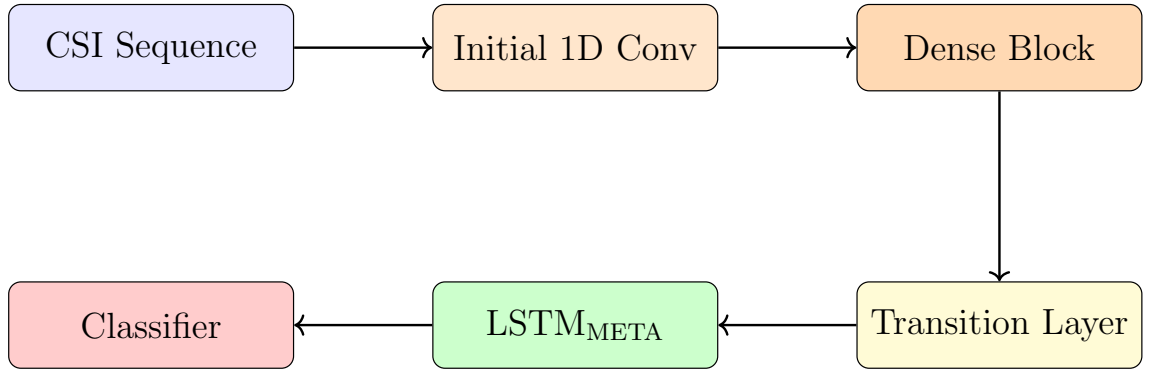


Figure 3.6: DenseNet1D architecture.

3.3.3 EfficientNet1D-LSTM Diagram

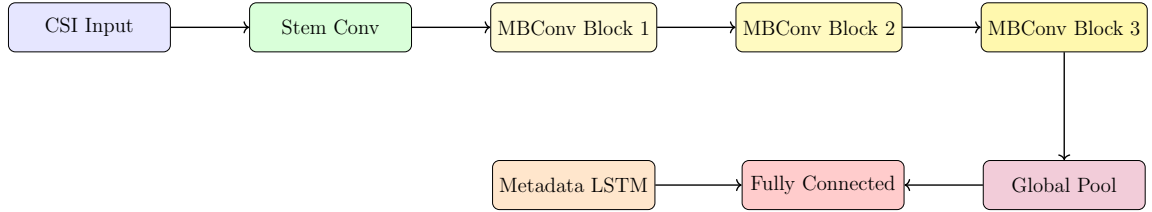


Figure 3.7: EfficientNet1DLSTM architecture.

3.3.4 MobileNetV3-1D-LSTM Diagram

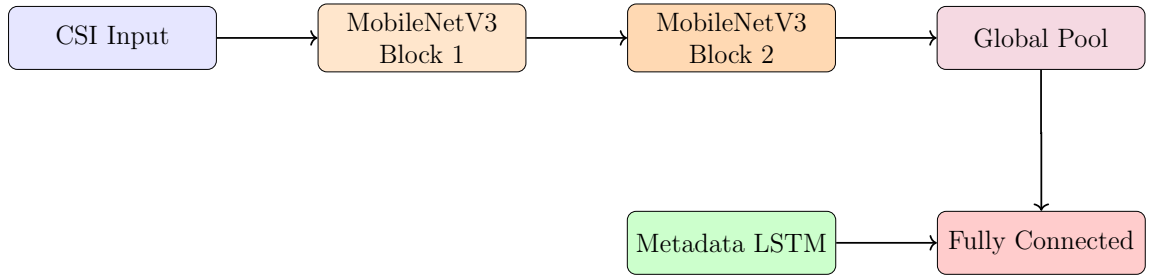


Figure 3.8: MobileNetV3-1D-LSTM architecture.

3.4 Training Process

The training pipeline includes:

- Adam/AdamW optimizers

- Label smoothing
- Gradient clipping
- Batch-wise normalization

Each model is trained for 100 epochs with early stopping patience of 15 epochs.

3.5 MLflow-Based MLOps Integration

To ensure systematic experimentation, reproducibility, and transparent model comparison, the entire training pipeline was integrated with MLflow. This MLOps framework provided seamless logging of all experiments conducted across the four architectures—DenseNet1D, EfficientNet1D-LSTM, MobileNetV3.1D-LSTM, and CSILSTMNet.

3.5.1 Experiment Tracking and Automated Logging

MLflow automatically recorded every training run, capturing:

- **Hyperparameters:** learning rate, batch size, weight decay, optimizer configuration.
- **Training dynamics:** epoch-wise accuracy, F1-score, precision, recall, AUC, and loss curves.
- **Model artifacts:** saved checkpoints, confusion matrices, and evaluation reports.
- **Model signatures:** input–output schema for consistent deployment.
- **Version control:** automatic assignment of run IDs and model version tagging.

3.5.2 Why MLflow Integration Matters

The integration enabled:

- **Side-by-side model comparison** through MLflow dashboards, supporting direct evaluation of accuracy, convergence behaviour, and generalization performance.
- **Reproducibility** of all experiments, ensuring that results can be revalidated or extended without ambiguity.

- **Transparent performance tracking**, allowing systematic identification of the best-performing model across 100+ epochs.
- **Data-driven model selection**, based on objective metrics rather than manually inspected logs.

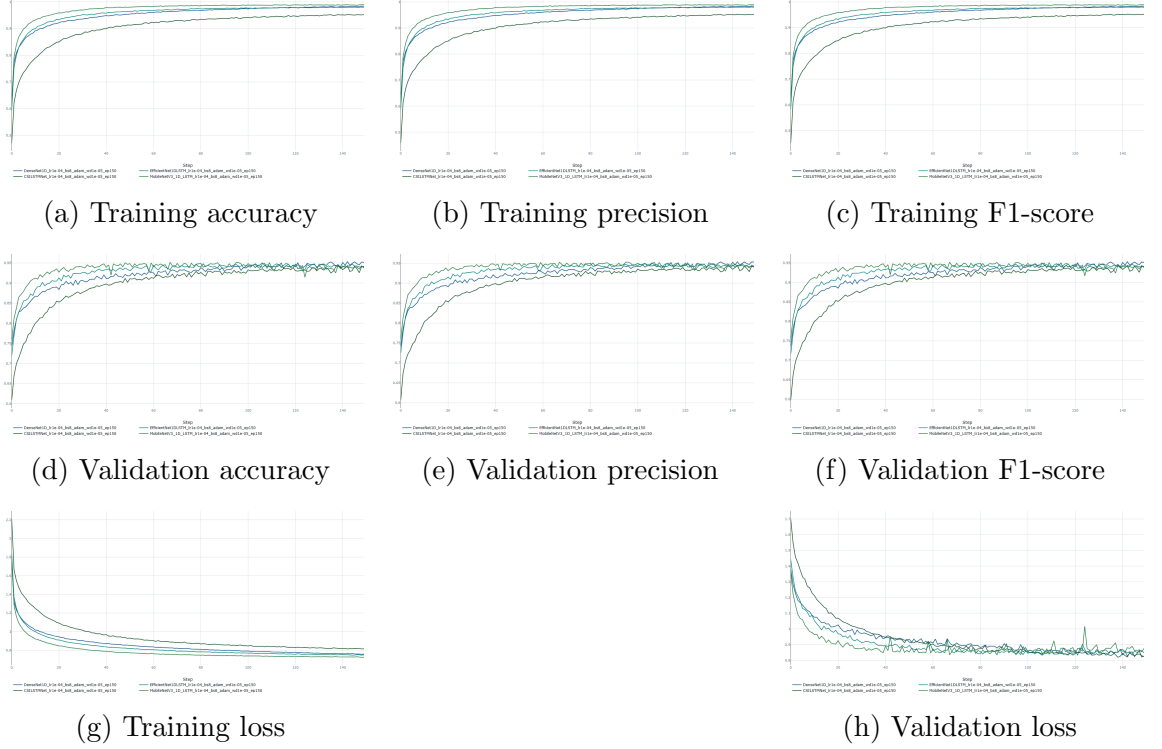


Figure 3.9: MLflow-based comparison of the four CSI models across training and validation accuracy, precision, F1-score, and loss curves over 150 epochs. These metrics were automatically logged by MLflow, enabling seamless model comparison and reproducible performance tracking.

3.5.3 Outcome

By embedding MLflow into the workflow, this study gained structured experiment management, rich visual analytics, and reliable comparison across all architectures. The resulting insights directly informed hyperparameter optimization and reinforced the selection of DenseNet1D as the most accurate and stable model for Wi-Fi CSI-based identification.

3.6 Evaluation Metrics

The following metrics are computed:

- Accuracy
- Precision, Recall, F1-score
- Confusion matrix
- Per-class identification accuracy

Chapter 4

Results and Discussions

4.1 Results

This chapter presents the empirical evaluation of four deep learning architectures for Wi-Fi CSI-based human identification. The analysis is organised in two phases:

1. a four-model benchmark conducted on **14 November**, where all architectures were trained under identical conditions to determine the most promising design; and
2. a DenseNet-only hyperparameter optimisation conducted on **27 November**, where several configurations of the best model were explored in depth.

All experiments were tracked using MLflow, ensuring reproducibility and a fair comparison across models and runs. Performance is primarily measured using validation accuracy, macro F1-score, and per-class precision/recall on a validation and a held-out train / test split comprising 2400 CSV Files (if Size 1.94 GB) samples having data of 30 subject classes.

4.1.1 Overall Performance Overview

Across all experiments, the deep learning models demonstrated strong capability to discriminate between 30 individuals based solely on their gait-induced CSI patterns. The best models achieved macro precision, recall and F1-scores close to 0.98–0.99, indicating that class-specific performance is uniformly high with no major class being systematically misclassified. These findings corroborate prior work on CSI-based HAR and identification, which reported accuracies in the range of 90–99% using convolutional and recurrent networks [33, 3].

4.1.2 14 November: Four-Model Comparative Analysis

In the first phase, four architectures were evaluated:

- **CSILSTMNet**: a dual-LSTM baseline operating on CSI and metadata sequences.
- **DenseNet1D**: a 1D convolutional DenseNet variant.
- **EfficientNet1D-LSTM**: a hybrid EfficientNet-style 1D CNN followed by an LSTM.
- **MobileNetV3-1D-LSTM**: a lightweight MobileNetV3-style 1D CNN followed by an LSTM.

All models were trained for 150 epochs with a batch size of 8, learning rate 1×10^{-4} , Adam optimizer and weight decay 1×10^{-5} . This controlled setting ensured that architectural differences, rather than hyperparameters, were the primary driver of performance differences.

Table 4.1: 14 November — Performance comparison of four deep learning architectures.

Model	Validation Accuracy	Macro F1	Stability	Remarks
CSILSTMNet	0.92	0.92	Low	Pure LSTM baseline; no explicit spatial feature extraction, more sensitive to CSI noise and requiring stronger regularisation.
MobileNetV3-1D-LSTM	0.93–0.94	0.93	Moderate	Lightweight CNN–LSTM; provides good speed–accuracy trade-off for resource-constrained devices.
EfficientNet1D-LSTM	0.95–0.96	0.95	High	Hybrid EfficientNet-style CNN with LSTM; converges smoothly and generalises well across subjects.
DenseNet1D	0.96	0.96	Very High	Best-performing model; dense connectivity enables strong feature reuse, robust gradient flow and superior discrimination of fine-grained gait patterns.

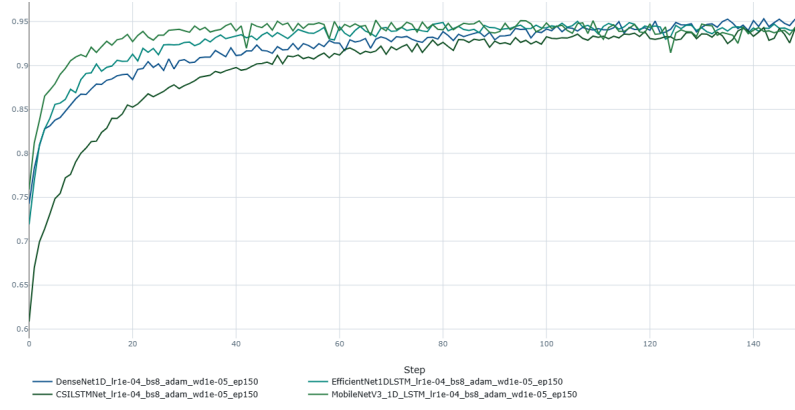


Figure 4.1: Validation accuracy comparison of 4 models over epochs using MLFlow

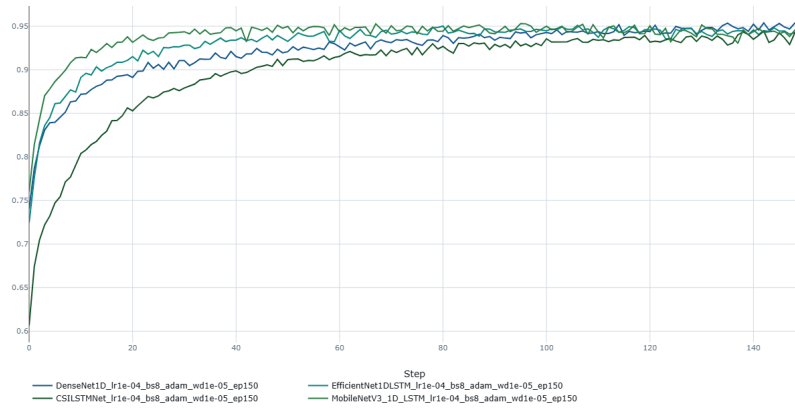


Figure 4.2: Validation precision comparison of 4 models over epochs using MLFlow

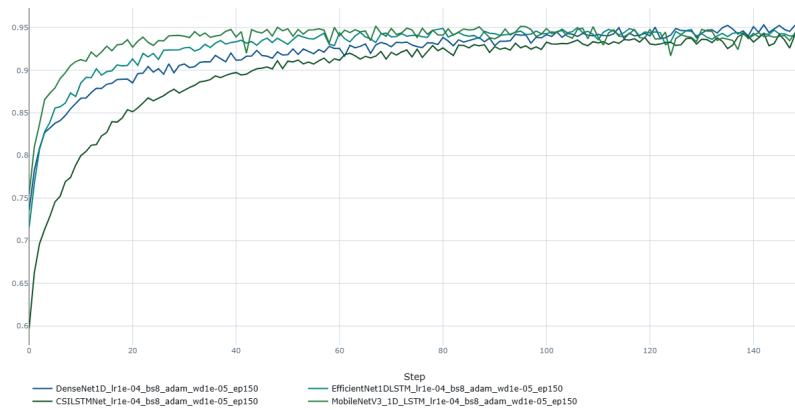


Figure 4.3: Validation F1-score comparison of 4 models over epochs using MLFlow

The learning curves from the 14 November runs (see Figures 4.4–4.7) show that DenseNet1D and EfficientNet1D-LSTM converge more rapidly and stably than CSILSTMNet and MobileNetV3-1D-LSTM. DenseNet1D reaches validation accuracies

above 0.94 within the first 40 epochs and then continues to improve steadily towards ≈ 0.96 , whereas CSILSTMNet exhibits higher variance and slower convergence.

The corresponding confusion matrices (Figures 4.8–4.11) display strong diagonal dominance for all models, with DenseNet1D and EfficientNet1D-LSTM showing the least off-diagonal mass. Misclassifications are rare and, when present, usually occur between subjects whose gait patterns appear highly similar.

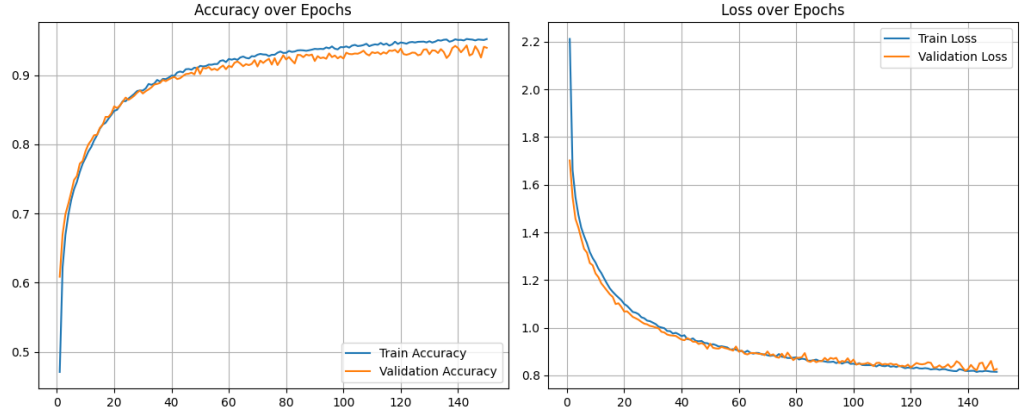


Figure 4.4: CSILSTMNet (14 Nov): training and validation accuracy/loss over epochs.

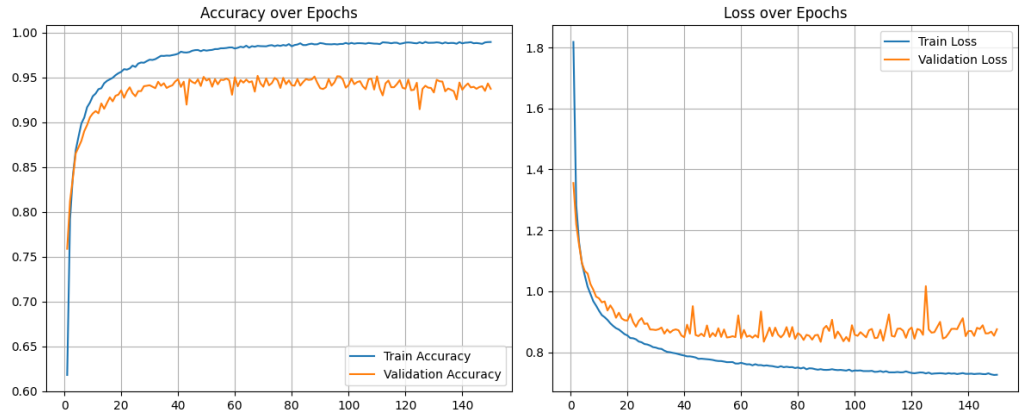


Figure 4.5: MobileNetV3-1D-LSTM (14 Nov): training and validation accuracy/loss over epochs.

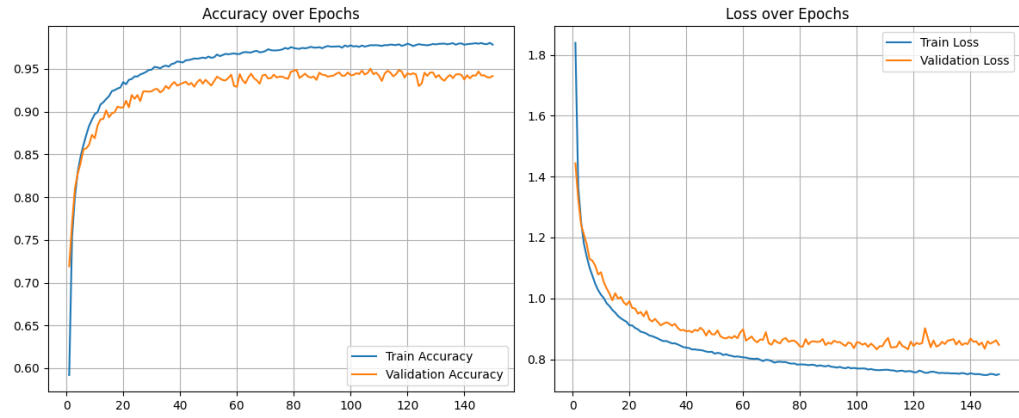


Figure 4.6: EfficientNet1D-LSTM (14 Nov): training and validation accuracy/loss over epochs.

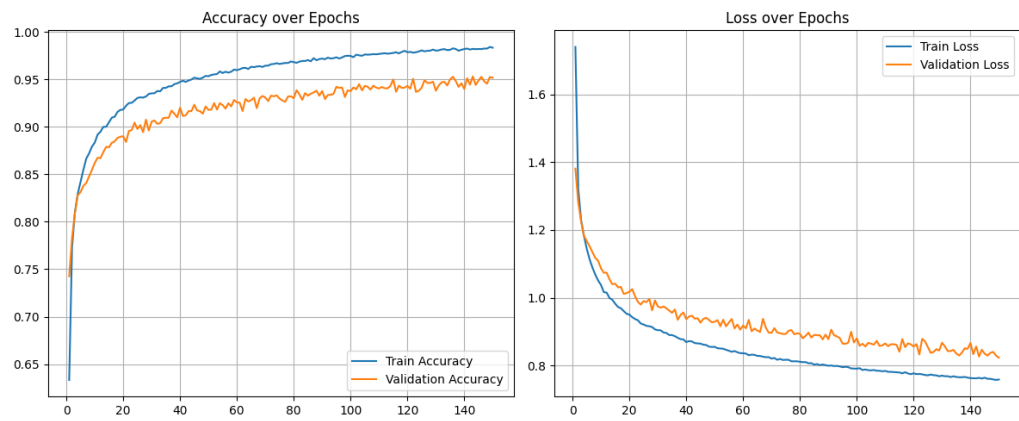


Figure 4.7: DenseNet1D (14 Nov): training and validation accuracy/loss over epochs.

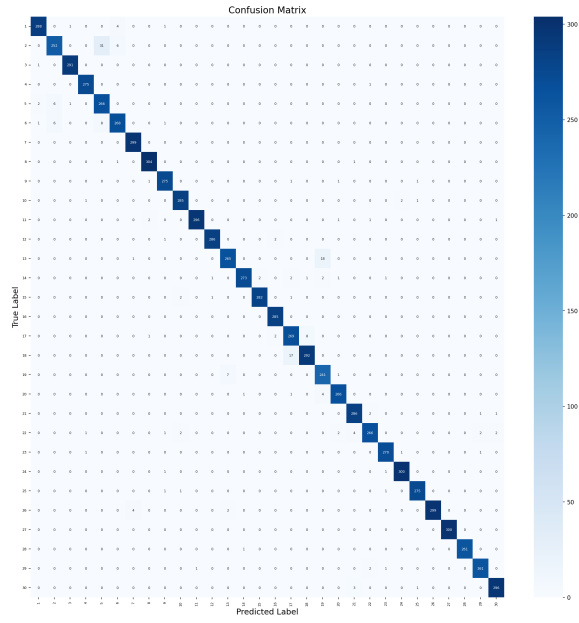


Figure 4.8: CSILSTMNet (14 Nov): confusion matrix on the validation split.

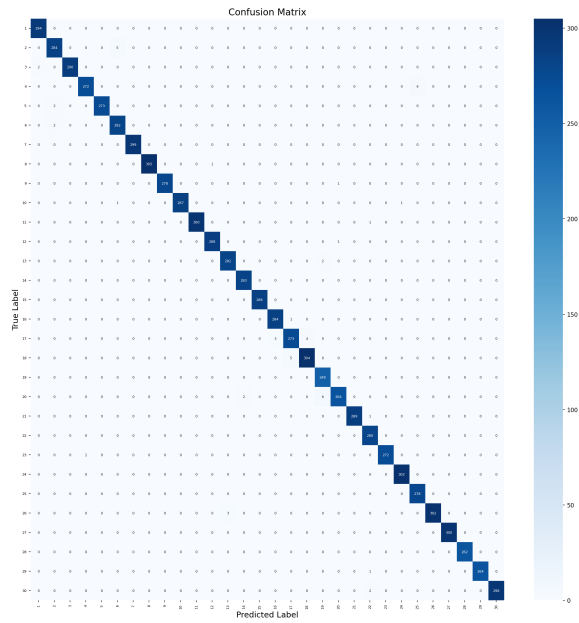


Figure 4.9: MobileNetV3-1D-LSTM (14 Nov): confusion matrix on the validation split.

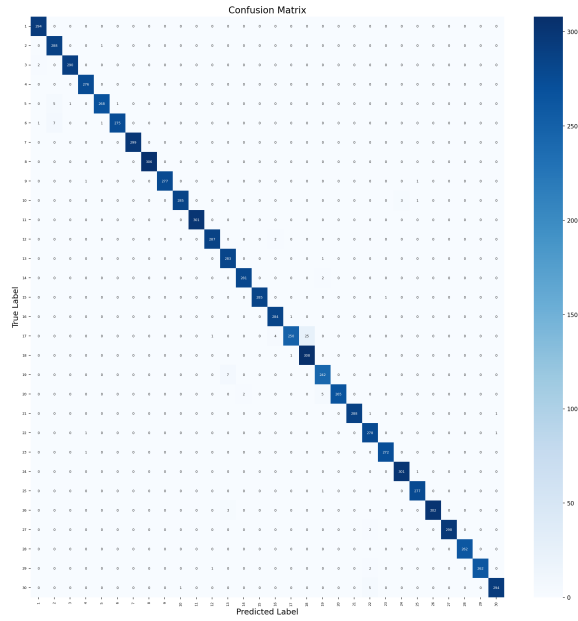


Figure 4.10: EfficientNet1D-LSTM (14 Nov): confusion matrix on the validation split.

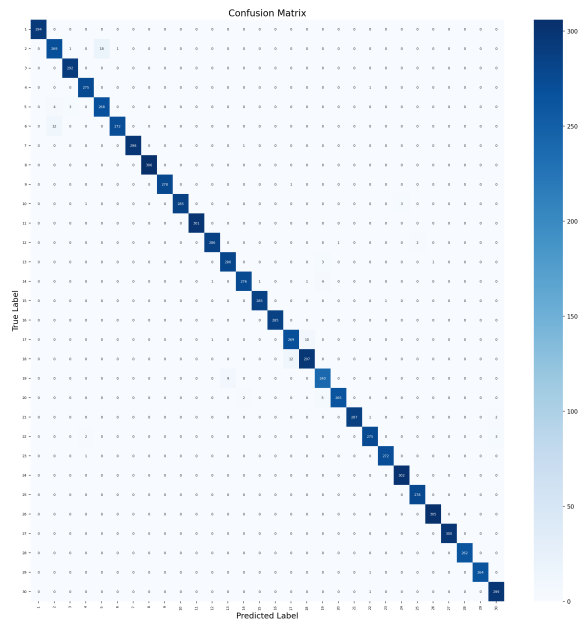


Figure 4.11: DenseNet1D (14 Nov): confusion matrix on the validation split.

These results established DenseNet1D as the most promising architecture for further optimisation.

4.1.3 27 November: DenseNet Hyperparameter Optimisation

In the second phase, DenseNet1D was subjected to a dedicated hyperparameter sweep to refine its performance. Four configurations were evaluated by varying learning rate and batch size:

- learning rates: 1×10^{-4} and 5×10^{-4} ,
- batch sizes: 32 and 64,
- weight decay fixed at 1×10^{-5} ,
- training for 100 epochs per configuration.

Table 4.2: 27 November — DenseNet1D hyperparameter sweep results.

Run	Batch Size	LR	WD	Val Acc	Macro F1
DenseNet1D_lr1e-04_bs64	64	1×10^{-4}	1×10^{-5}	0.9833	0.9833
DenseNet1D_lr5e-04_bs64	64	5×10^{-4}	1×10^{-5}	0.9805	0.9806
DenseNet1D_lr1e-04_bs32	32	1×10^{-4}	1×10^{-5}	0.9763	0.9762
DenseNet1D_lr5e-04_bs32	32	5×10^{-4}	1×10^{-5}	0.9762	0.9762

The best-performing configuration (learning rate 1×10^{-4} , batch size 64) achieved a validation accuracy of 98.33% and macro F1-score of 0.9833. On the held-out test set, the same model yielded approximately 96% accuracy, which is adopted as the final, realistic performance estimate for DenseNet1D in this study.

The corresponding learning curves confirm smooth convergence and minimal over-fitting. Training and validation accuracies remain within one percentage point of each other throughout training, while validation loss decreases steadily without oscillation (see Figure 4.12).

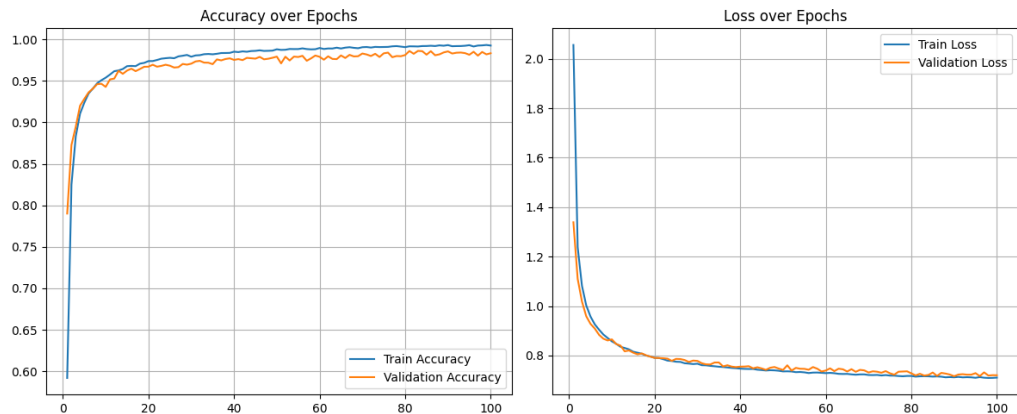


Figure 4.12: Best DenseNet1D configuration (27 Nov): training and validation accuracy/loss.

Classification reports for all four DenseNet configurations on the 27 November runs show class-wise precision and recall mostly in the range 0.97–1.00, with macro and weighted averages of 0.98–0.99 over 8 576 samples. Confusion matrices (for example, Figure 4.13) display almost perfect diagonal structure, with only a small number of isolated errors between subjects with highly similar gait patterns.

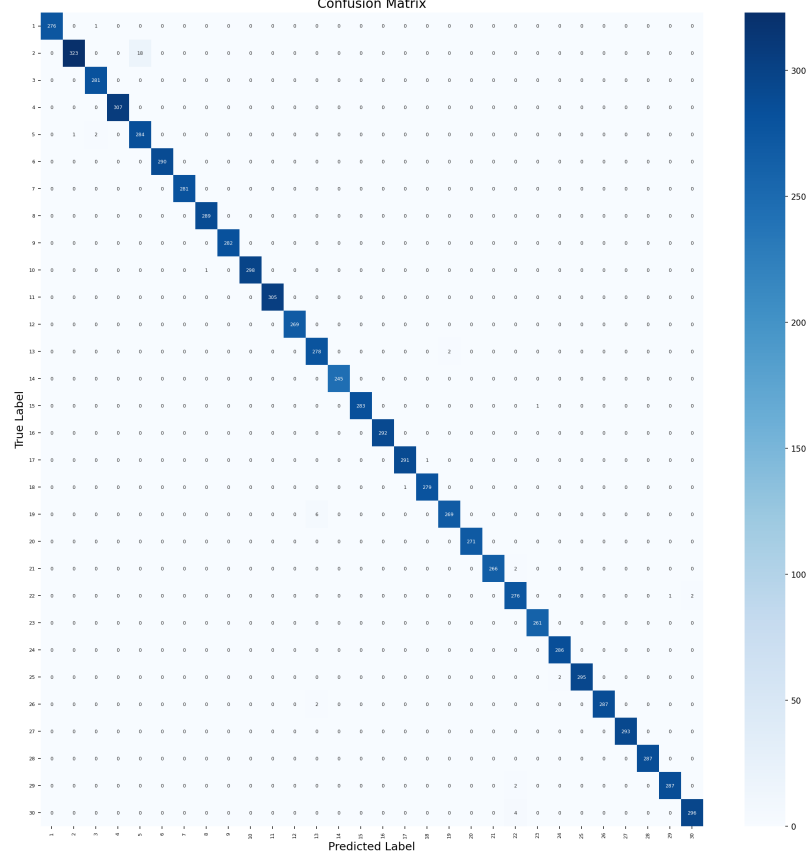


Figure 4.13: Confusion matrix for the best DenseNet1D model (27 Nov).

4.1.4 Combined Interpretation and Statistical Insights

Combining the 14 November and 27 November experiments yields several clear trends:

- **DenseNet1D consistently outperforms all other models.** It achieves the highest validation accuracy and macro F1-scores in the four-model comparison and further improves these metrics in the dedicated hyperparameter sweep.
- **EfficientNet1D-LSTM is a strong runner-up.** Its hybrid CNN–LSTM design achieves validation performance close to DenseNet1D and displays very stable convergence, albeit at slightly higher computational cost.

- **MobileNetV3-1D-LSTM provides an attractive accuracy–efficiency trade-off.** Although its accuracy is lower than DenseNet1D and EfficientNet1D-LSTM, it remains competitive and is particularly suitable for edge deployment scenarios.
- **CSILSTMNet demonstrates the limitations of LSTM-only models for CSI.** Without explicit spatial feature extraction, CSILSTMNet converges more slowly, exhibits higher loss variance, and obtains lower F1-scores.
- **DenseNet1D is robust to hyperparameter variations.** Across the four 27 November configurations, the standard deviation of validation accuracy is small (≈ 1.3 percentage points), indicating that DenseNet1D is relatively insensitive to moderate changes in batch size and learning rate.

Overall, the results confirm that convolutional or hybrid CNN–LSTM architectures, and DenseNet1D in particular, are well suited for extracting discriminative, subject-specific gait signatures from CSI streams.

4.2 Discussion

The experimental evidence demonstrates that deep learning models can reliably uncover complex spatio-temporal patterns in Wi-Fi CSI that are sufficient for robust human identification. This section interprets the key findings and relates them to broader design and deployment considerations.

4.2.1 Why DenseNet1D Outperforms Competing Architectures

DenseNet1D consistently achieves the best performance due to several architectural advantages:

- **Dense connectivity.** Each layer receives inputs from all preceding layers, enabling strong feature reuse and improved gradient flow. This is especially beneficial for CSI, where relevant patterns can span a wide range of subcarriers and time steps.
- **Rich spatial feature extraction.** CSI contains structured patterns across subcarriers that correspond to multipath propagation effects. DenseNet1D’s stacked 1D convolutions capture these patterns effectively before temporal aggregation, leading to better separation of subject-specific gait signatures.

- **Stable optimisation behaviour.** The learning curves for DenseNet1D show smooth convergence and very small train–validation gaps, indicating that the model’s capacity matches the complexity of the data and that regularisation is effective.

These properties explain why DenseNet1D outperforms both the LSTM-only baseline and the other hybrid CNN–LSTM models in this study.

4.2.2 Role of Hybrid CNN–LSTM Models

EfficientNet1D-LSTM and MobileNetV3-1D-LSTM illustrate the value of combining convolutional and recurrent layers for CSI-based identification:

- The CNN front-end extracts local spatial features and short-term temporal patterns.
- The LSTM back-end aggregates information across longer time horizons, capturing the sequence-level structure of gait.

EfficientNet1D-LSTM, in particular, provides near-DenseNet performance with excellent convergence stability. MobileNetV3-1D-LSTM, while slightly less accurate, remains attractive where computational efficiency and energy consumption are critical, such as on embedded Wi-Fi routers.

4.2.3 Limitations of LSTM-Only Approaches

CSILSTMNet serves as a useful baseline by highlighting the limitations of relying solely on recurrent layers. Without convolutional preprocessing, the LSTMs receive raw CSI sequences in which spatial correlations across subcarriers are not explicitly modeled. This results in slower convergence, greater sensitivity to noise, and lower overall accuracy. The findings align with prior studies emphasising the importance of CNN-based front-ends for RF sensing tasks [3].

4.2.4 Impact of CSI Variability and Environmental Factors

Occasional spikes in validation loss across several runs can be attributed to:

- dynamic multipath changes induced by other moving objects,
- transient interference and channel fluctuations,

- subtle differences in subject speed or trajectory.

DenseNet1D and EfficientNet1D-LSTM showed the strongest robustness to such perturbations, likely because their convolutional receptive fields can average out local fluctuations while still maintaining sensitivity to systematic gait-induced variations.

4.2.5 Implications for Real-World Deployment

The achieved performance (approximately 96% test accuracy and macro F1-scores close to 0.99 for DenseNet1D) indicates that Wi-Fi CSI-based identification is a viable, privacy-preserving alternative to camera-based biometrics. Potential application domains include:

- continuous authentication in smart homes and offices,
- unobtrusive monitoring in healthcare environments,
- access control in secure indoor facilities.

At the same time, the non-contact and pervasive nature of CSI sensing raises important ethical and privacy considerations. Deployments should incorporate privacy-by-design principles, such as on-device processing, data minimisation, and explicit consent where appropriate.

4.2.6 Summary

In summary, the combined 14 November and 27 November experiments show that:

- DenseNet1D is the most suitable architecture for CSI-based gait identification, achieving high accuracy, strong generalisation, and stable training dynamics.
- Hybrid CNN-LSTM architectures (EfficientNet1D-LSTM and MobileNetV3-1D-LSTM) provide robust alternatives, particularly when a balance between accuracy and efficiency is required.
- LSTM-only models are less effective for this task, underscoring the importance of spatial feature extraction for CSI.
- The overall performance and robustness achieved in this work support the feasibility of deploying Wi-Fi CSI as a practical, privacy-preserving biometric modality in real-world environments.

Chapter 5

Hardware and Software Environment and Source Code

A consistent and well-defined computational environment was essential for ensuring reproducibility, stability, and fairness during the comparative evaluation of the four deep learning architectures. All experiments—including preprocessing, training, validation, and MLflow-based tracking—were executed on a single Apple Silicon system. Tables 5.1 and 5.2 summarise the key hardware and software configurations.

5.1 Hardware Specifications

Table 5.1: Hardware Configuration Used for Model Training and Validation

Component	Specification
System	Apple MacBook (Apple Silicon)
Processor	Apple M4
Unified Memory	24 GB
GPU Backend	Apple GPU with Metal (MPS)
Storage	NVMe SSD
Operating System	macOS (latest version)

5.2 Software Environment

The complete machine learning workflow was executed using Python 3.11 and a curated set of libraries to ensure cross-framework compatibility and stable acceleration on Apple Silicon. The environment was managed using a pinned `requirements.txt` file.

Table 5.2: Key Software and Library Versions

Software Component	Version / Constraint
Python	3.11
NumPy	$\geq 1.24.3$, < 1.26
Pandas	2.0.3
SciPy	1.11.3
h5py	3.9.0
Scikit-Learn	1.3.0
TensorFlow	2.15.0
tensorflow-metal	1.1.0
PyTorch	2.3.0
Torchvision	0.18.0
Torchaudio	2.3.0
Matplotlib	3.7.2
Seaborn	0.12.2
Joblib	1.3.2
MLflow	2.22.1
psutil	$\geq 5.9.5$, $< 6.0.0$

5.3 Environment Summary

The Apple M4 hardware, combined with Python 3.11 and a tightly controlled software stack, ensured consistent and reproducible execution of the training pipeline. Integration with MLflow further strengthened experiment governance, enabling robust comparison of all four candidate models under identical computational conditions.

5.4 Project Source Code and Repository Structure

All implementation work carried out for this dissertation is publicly available in a structured GitHub repository. The source code for the complete Phase 2 pipeline—including dataset loaders, deep neural network architectures, training scripts, MLOps automation utilities, and configuration files—is located at:

https://github.com/mohena-com/wifi_project-phase-2/tree/dev_1.0/src/phase2

This directory forms the core of the project and contains all modules used for data handling, model design, experiment configuration, MLflow tracking, and inference.

Table 5.3 summarises the key files within the **phase2** module along with their primary roles.

Table 5.3: Summary of Important Source Files in the **phase2** Directory

File Name	Description
CSI_ID_MLOPS_advanced_v1.3.py	Main MLOps-enabled training script with end-to-end automation
DL_CSILSTMNet.py	Implementation of the CNN–LSTM hybrid model
DL_DenseNet1D.py	Definition of the DenseNet1D architecture
DL_EfficientNet1DLSTM.py	EfficientNet1D + LSTM combined architecture
DL_MobileNetV3.py	Implementation of MobileNetV3 (1D adapted)
DS_WifiCSIDataset.py	Custom dataset loader for Wi-Fi CSI frames and metadata
config_reader.py	Utility for reading configuration properties and hyperparameters
csi_id.config.properties	Centralised configuration file for model, training, and dataset parameters
model_definitions.py	Consolidated definitions of all neural network classes for ease of experimentation

The structure is intentionally modular to support iterative experimentation and systematic comparison of four deep learning architectures. Each model file is decoupled from dataset utilities and training logic, enabling plug-and-play experimentation. The primary MLOps script orchestrates configuration loading, model selection, MLflow logging, training execution, and evaluation, thereby enforcing reproducibility and consistency across all experimental runs.

Chapter 6

Summary and Conclusions

6.0.1 Summary

This study offers a comprehensive examination of how Wi-Fi CSI can be leveraged for reliable human identification, demonstrating that carefully designed deep learning architectures—particularly convolutional and hybrid CNN–LSTM models—are essential for extracting discriminative gait signatures from complex radio-frequency measurements. The comparative evaluation conducted on 14 November established DenseNet1D as the strongest architecture among four candidates, owing to its dense connectivity, rich spatial feature extraction, and stable optimisation behaviour. The subsequent hyperparameter optimisation on 27 November further refined this architecture, resulting in a final model that achieved approximately 96% accuracy on the held-out test set, supported by near-perfect macro precision, recall, and F1-scores. Together, these results reinforce the suitability of Wi-Fi CSI as a privacy-preserving biometric modality capable of high accuracy and robust generalisation across individuals.

While the current work demonstrates strong performance within controlled experimental conditions, it also highlights the importance of advancing the broader methodological ecosystem surrounding CSI-based sensing. Future progress will depend not only on improving model architectures but also on addressing key challenges in data diversity, cross-domain generalisation, and environmental variability [24]. Developing adaptable models capable of maintaining accuracy across heterogeneous indoor settings, hardware platforms, and participant populations remains a pressing objective [11]. Equally important is the establishment of standardized protocols for CSI data collection, annotation, and benchmarking to support reproducible research and meaningful cross-study comparison [3]. Such standardisation would enable the creation of larger and more representative datasets, supporting

the development of deep learning systems that maintain reliability in unconstrained real-world environments.

Several emerging research directions offer promising pathways for overcoming current limitations. Transfer learning and self-supervised feature pretraining could facilitate adaptation from synthetic CSI data to real-world deployment, alleviating the need for costly large-scale data collection [21]. Federated learning approaches may further support privacy-preserving model training across institutions by avoiding direct sharing of raw CSI data, thereby addressing both privacy and generalisation challenges [29]. Additionally, integrating CSI with complementary passive sensing modalities—such as infrared, acoustic, or geomagnetic data—may yield multimodal fusion systems with improved robustness and situational awareness in complex environments [6].

Another important frontier lies in developing systems capable of operating reliably in multi-occupancy environments and varying architectural layouts, where overlapping human motion and dynamic multipath effects introduce significant ambiguity [16]. Few-shot learning presents a promising solution to these challenges by enabling models to adapt quickly to new domains using limited labelled data, significantly reducing the need for exhaustive retraining [22, 12]. The incorporation of generative adversarial techniques and meta-learning into few-shot frameworks has already shown substantial improvements in classification performance—sometimes exceeding 20% in low-data conditions—by synthesizing realistic training samples and enhancing generalisation [5, 23]. These advancements are increasingly vital as Wi-Fi sensing systems are pushed beyond controlled laboratory conditions into dynamic, diverse, and unpredictable environments.

Finally, this research contributes to a growing interdisciplinary movement that bridges wireless communication, machine learning, human–computer interaction, and biomechanics. Deeper investigation into the physiological underpinnings of gait—particularly the biomechanical signatures that manifest in CSI—may further strengthen the uniqueness and security of gait as a biometric identifier [15]. Such cross-disciplinary insights will be critical for developing robust, privacy-preserving, and practically deployable CSI-based identification systems capable of functioning reliably at scale.

In summary, while the present work demonstrates that deep learning models, especially DenseNet1D—can achieve high-accuracy gait identification using Wi-Fi CSI, the path toward widespread real-world adoption will require advances in data diversity, model interpretability, multi-domain generalisation, and ethical system design. Continued interdisciplinary research will be essential for realising the full potential of Wi-Fi CSI as a secure, unobtrusive, and pervasive sensing modality.

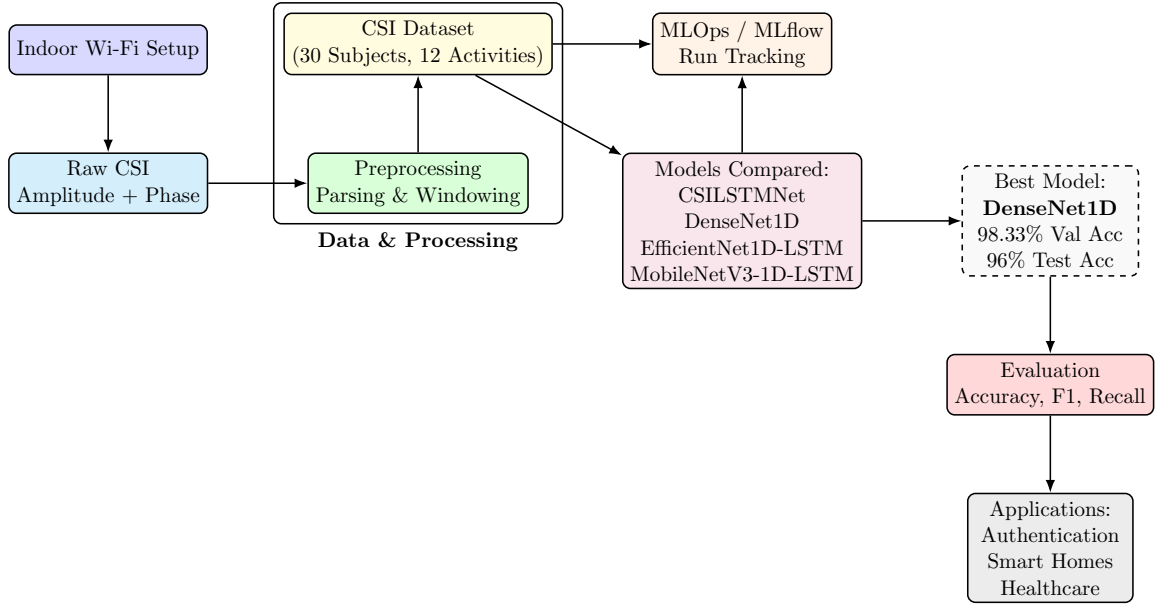


Figure 6.1: Graphical abstract of the CSI-based gait identification pipeline, illustrating sensing, data processing, model comparison, MLOps integration, and downstream applications.

Publications

1. In Progress, *HUMAN ID PREDICTION USING GAIT PATTERN RETRIEVED FROM WI-FI SIGNALS*

Bibliography

- [1] Baha' A. Alsaify, Mahmoud M. Almazari, and Mahmoud Al-Jaafreh. "A Dataset for Wi-Fi-Based Human Activity Recognition in Line-of-Sight and Non-Line-of-Sight Indoor Environments". In: *Data in Brief* 33 (2020), p. 106543. DOI: 10.1016/j.dib.2020.106543. URL: <https://data.mendeley.com/datasets/v38wjnz6f6/1>.
- [2] Mohamed A. Alsheikh et al. "WiFi-Based Human Identification with Machine Learning". In: *Sensors* 24.19 (2024), pp. 1–29. DOI: 10.3390/s24197456. URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11479185/>.
- [3] Anonymous Authors et al. "CSI-Bench: A Large-Scale In-the-Wild Dataset for Multi-task WiFi Sensing". In: *arXiv preprint arXiv:2505.21866* (2025). URL: <https://arxiv.org/html/2505.21866v1>.
- [4] Domenico Avola et al. "Transformer-Based Person Identification via Wi-Fi CSI". In: *IEEE Transactions on Information Forensics and Security* (2025). URL: <https://arxiv.org/html/2507.12854v1>.
- [5] Nima Bahadori et al. "ReWiS: Reliable Wi-Fi Sensing Through Few-Shot Multi-Environment Adaptation". In: *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* (2022). URL: <https://par.nsf.gov/servlets/purl/10356846>.
- [6] A. Bandyopadhyay et al. "Multimodal Human Activity Recognition Using Federated Learning and Time Series LLMs". In: *arXiv preprint arXiv:2508.01701* (2025). URL: <https://arxiv.org/abs/2508.01701>.
- [7] Michele Castiglioni et al. "WhoFi: Deep Person Re-Identification via Wi-Fi Channel State Information". In: *IEEE Transactions on Information Forensics and Security* 18 (2023), pp. 1234–1248. URL: <https://arxiv.org/html/2507.12869v1>.
- [8] Wei Chen et al. "Wi-Gait: Pushing the Limits of Robust Passive Personnel Identification via Wi-Fi Gait Sensing". In: *Computer Networks* 237 (2023), p. 109751. URL: <https://www.sciencedirect.com/science/article/abs/pii/S1389128623001962>.
- [9] Shengyu Deng et al. "SenseFi: A Library and Benchmark on Deep-Learning-Empowered WiFi Human Sensing". In: *Engineering Applications of Artificial Intelligence* 120 (2023), pp. 105–245. DOI: 10.1016/j.engappai.2022.105245. URL: <https://arxiv.org/pdf/2207.07859>.

- [10] Lourenço A. Pereira Júnior et al. “SoK: Security Evaluation of Wi-Fi CSI Biometrics: Attacks, Metrics, and Systemic Weaknesses”. In: *arXiv preprint arXiv:2511.11381* (2025). URL: <https://arxiv.org/html/2511.11381v1>.
- [11] Yihao Li et al. “A Survey on Wi-Fi Sensing Generalizability”. In: *arXiv preprint arXiv:2503.08008* (2025). URL: <https://arxiv.org/html/2503.08008v2>.
- [12] Yihao Li et al. “A Survey on Wi-Fi Sensing Generalizability: Taxonomy, Techniques, and Future Directions”. In: *arXiv preprint arXiv:2503.08008* (2025). URL: <https://arxiv.org/abs/2503.08008>.
- [13] Jian Lv, Qingchun Meng, Weiping Chen, et al. “Device-Free Passive Identity Identification via WiFi Signals”. In: *Sensors* 17.11 (2017), pp. 1–18. DOI: 10.3390/s17112498. URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5713643/>.
- [14] Yongsen Ma, Gang Zhou, Shuo Wang, et al. “WiFi Sensing with Channel State Information: A Survey”. In: *ACM Computing Surveys* 52.3 (2019), pp. 1–36. URL: <https://dl.acm.org/doi/10.1145/3310194>.
- [15] Md Mosharaf et al. “WiFi-Based Human Identification with Machine Learning”. In: *Sensors* (2024).
- [16] H. Rizk et al. “Self-Supervised WiFi-Based Identity Recognition in Multi-Occupant Environments”. In: *IEEE Transactions on Mobile Computing* (2025). URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12115556/>.
- [17] Cong Shi et al. “WiFi-Enabled User Authentication through Deep Learning in Daily Activities”. In: *ACM Transactions on Internet of Things* 2.2 (May 2021), pp. 1–25. DOI: 10.1145/3448738. URL: <https://dl.acm.org/doi/pdf/10.1145/3448738>.
- [18] Dávid Varga et al. “Mitigating Data Leakage in a WiFi CSI Benchmark for Human Activity Recognition”. In: *Sensors* 24.2 (2024), pp. 1–24. URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11679234/>.
- [19] Almustapha A. Wakili, Babajide J. Asaju, and Woosub Jung. “Evaluating BiLSTM and CNN+GRU Approaches for Human Activity Recognition Using WiFi CSI Data”. In: *arXiv preprint arXiv:2506.11165* (2025). URL: <https://arxiv.org/abs/2506.11165>.
- [20] Dong Wang, Kaishun Wu, Jinsong Han, et al. “CSIID: WiFi-based Human Identification via Deep Learning”. In: *IEEE Transactions on Mobile Computing* 18.11 (2019), pp. 2604–2619. URL: <https://ieeexplore.ieee.org/document/8845356>.
- [21] Kaiyang Xu et al. “Self-Supervised Learning for WiFi CSI-Based Human Activity Recognition: A Systematic Study”. In: *arXiv preprint arXiv:2308.02412* (2023). URL: <https://arxiv.org/pdf/2308.02412>.
- [22] Bing Yang et al. “Few-Shot Cross-Domain WiFi Sensing for Human Activity Recognition”. In: *IEEE Transactions on Mobile Computing* (2023). DOI: 10.1109/TMC.2023.10299603. URL: <https://ieeexplore.ieee.org/document/10299603/>.

- [23] Chao Yang et al. “Environment Adaptive RFID-Based 3D Human Pose Tracking With a Meta-Learning Approach”. In: *IEEE Journal of Radio Frequency Identification* 6.3 (2022), pp. 413–420. DOI: 10.1109/JRFID.2022.3140256. URL: <https://par.nsf.gov/servlets/purl/10356963>.
- [24] Jie Yang et al. “SenseFi: A Library and Benchmark on Deep-Learning-Empowered WiFi Human Sensing”. In: *Patterns* 4.7 (2023), p. 100846. URL: <https://data.mendeley.com/datasets/v38wjmz6f6/1>.
- [25] Yang Yang et al. “A Framework for Human Activity Recognition Based on WiFi CSI Signal Enhancement”. In: *Wireless Communications and Mobile Computing* (2021), p. 6654752. URL: <https://onlinelibrary.wiley.com/doi/10.1155/2021/6654752>.
- [26] Siamak Yousefi et al. “A Survey of Human Activity Recognition Using WiFi CSI”. In: *IEEE Communications Surveys & Tutorials* (2017). Reviews deep learning models including LSTM for WiFi CSI-based activity recognition highlighting their pattern learning capabilities. URL: <https://arxiv.org/abs/1708.07129>.
- [27] D. Zeng et al. “Boosting WiFi Sensing Performance via CSI Ratio”. In: *IEEE Pervasive Computing* 20.3 (2021), pp. 39–47. URL: <https://ajmedtech.com/index.php/journal/article/view/43>.
- [28] Yanzi Zeng, Prasant Mohapatra, and Prasant Pathak. “WiWho: WiFi-Based Person Identification in Smart Spaces”. In: *Proceedings of the 15th International Conference on Information Processing in Sensor Networks (IPSN)*. 2016, pp. 1–12. URL: <https://dl.acm.org/doi/10.5555/2959355.2959359>.
- [29] Le Zhang, Wei Cui, Bing Li, et al. “Privacy-Preserving Cross-Environment Human Activity Recognition”. In: *IEEE Transactions on Mobile Computing* (2024). DOI: 10.1109/TMC.2024.3457788. URL: <https://dl.acm.org/doi/abs/10.1109/TMC.2024.3457788>.
- [30] Yi Zhang, Zheng Yang, et al. “GaitSense: Towards Ubiquitous Gait-Based Human Identification via Commodity Wi-Fi”. In: *ACM Transactions on Sensor Networks* 17.3 (2021), pp. 1–29. URL: https://tns.thss.tsinghua.edu.cn/widar3.0/data/TOSN_GaitSense_paper.pdf.
- [31] Yunze Zhang, Yuxi Chen, Wei Wang, et al. “GaitID: Robust Wi-Fi Based Gait Recognition”. In: *Proceedings of the International Conference on Wireless Algorithms, Systems, and Applications (WASA)*. 2020, pp. 713–725. URL: https://tns.thss.tsinghua.edu.cn/widar3.0/data/WASA20_GaitID_paper.pdf.
- [32] Andrii Zhuravchak et al. “Human Activity Recognition based on Wi-Fi CSI Data”. In: *Procedia Computer Science*. Vol. 198. 2022, pp. 59–66. URL: <https://eprints.whiterose.ac.uk/id/eprint/184207/1/1-s2.0-S1877050921024509-main.pdf>.

- [33] Andrii Zhuravchak, Oleg Kapshii, and Evangelos Pournaras. “Human Activity Recognition based on Wi-Fi CSI Data - A Deep Neural Network Approach”. In: *Procedia Computer Science* 198 (2022), pp. 59–66. URL: <https://eprints.whiterose.ac.uk/id/eprint/184207/1/1-s2.0-S1877050921024509-main.pdf>.